Comparative Evaluation of Classifiers for IoT Attack Type Detection

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Introduction

This project aims to categorise different types of IoT cybersecurity attacks using **supervised machine learning** techniques. IoT networks are typically lightweight, with limited processing power and storage capacity. These constraints often prevent the deployment of robust security measures, making IoT systems particularly vulnerable to various types of cyberattacks. Given the large volume of data generated by IoT devices, supervised learning presents a promising approach for enhancing attack detection, as it can learn to classify known attack types from labelled data. The objective of this project is to evaluate the extent to which supervised machine learning can effectively distinguish between different IoT attack types and to assess the responsiveness of the system in terms of processing speed.

The dataset **Cyber Attacks on Real-Time Internet of Things** (URL link) contains 123,117 rows and 85 columns. The target variable is attack_type, a categorical feature with the following 12 classes based on label indices: (0) *ARP_poisioning*, (1) *DDOS_Slowloris*, (2) *DOS_SYN_Hping*, (3) *MQTT_Publish*, (4) *Metasploit_Brute_Force_SSH*, (5) *NMAP_FIN_SCAN*, (6) *NMAP_OS_DETECTION*, (7) *NMAP_TCP_scan*, (8) *NMAP_UDP_SCAN*, (9) *NMAP_XMAS_TREE_SCAN*, (10) *Thing_Speak*, and (11) *Wipro_bulb*. *MQTT_Publish*, *Thing_Speak*, and *Wipro_Bulb_Dataset* are considered normal traffic, while the rest are attack categories. The features of this dataset can be broadly grouped into the following categories: (1) network and protocol information, (2) flow duration and inter-arrival time, (3) packet counts and transmission rates, (4) payload and header sizes, and (5) TCP flag counts. Among all features, only protocol and service are categorical; all others are numerical.

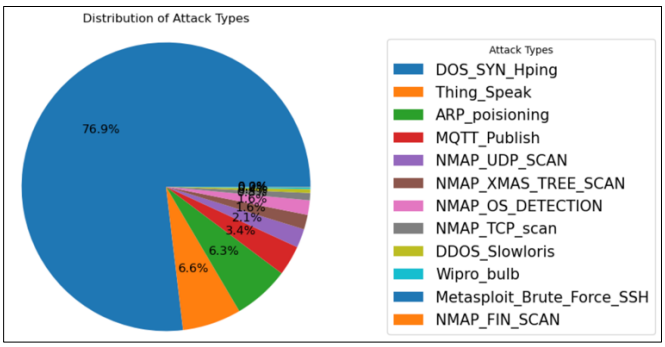

Fig1. Distribution of Attack Types

Literature Review

Prior research indicates that Random Forest performs well in IoT contexts due to its scalability, accuracy, and prediction speed (Hussain et al., 2020). While Decision Tree, Random Forest, and Artificial Neural Networks (ANN) can all achieve ~99% accuracy with large datasets, Random Forest is particularly effective for DoS attack detection (Hasan et al., 2019). However, in multiclass classification tasks, K-Nearest Neighbors (KNN) has been reported to outperform Random Forest by around 4% (Churcher et al., 2021).

Methods

1. Preprocessing

1.1 Missing Values

The service feature contains missing values represented as '-'. These were retained, as they account for 83.55% of the dataset and are closely associated with the DOS SYN Hping class. The service feature was retained as the Chi-square test showed a strong association with attack type (p-value = 0.0).

1.2 Removal of Rare Categories

To reduce noise and improve model performance, instances from extremely rare protocol and service categories were removed (122,816 records remained after removed).

- **Protocol:** Rows with ICMP protocol (only 0.05% of the data) were excluded.
- **Service:** Rare services including NTP, DHCP, IRC, SSH, and radius (together comprising 0.2% of the data) were removed.
- **Rare attack types** like *Metasploit_Brute_Force_SSH* and *NMAP_FIN_SCAN* were retained due to their distinctive characteristics.

2. Feature Engineering

2.1 Feature Construction

Most features are already statistical summaries (e.g., min, max, avg), so additional construction was limited to TCP flag-based ratios:

- **SYN ACK ratio:** High values may indicate SYN flood or scanning attacks.
- **RST ratio:** May indicate rejected or prematurely terminated connections.

2.2 Feature Selection

Highly correlated and low relevance features were removed.

- The ID column and port number columns (id.orig_p, id.resp_p) were removed as they offer no predictive value and have extremely high cardinality.

- Highly correlated features (correlation > 0.9) were grouped, and the one with the highest mutual information was retained. This reduced the feature set from 81 (after removing ID and port number columns) to 51.
- Features with mutual information scores below 0.1 were removed, resulting in 11 additional features being dropped.

2.3 Encoding

Since the categorical features proto (2 types) and service (5 types) have no ordinal relationship, they were one-hot encoded to represent each category as a separate binary feature. The target variable attack type was label encoded.

2.4 Feature Standardisation and Dimensionality Reduction

Right-skewed numerical features were log-transformed to reduce skewness, followed by standardisation of all numerical features. PCA was then applied, reducing the dimensionality from 40 to 7.

3. Imbalance Handling

The dataset was highly imbalanced. To address this, both random undersampling and SMOTE oversampling were tested across three models with both original and PCA-transformed features, resulting in 12 configurations.

4. Model

4.1 Model Selection

Three models were selected based on literature and to cover a bias-variance spectrum: **(1) Logistic Regression (LR), (2) Random Forest (RF), and (3) KNN.**

4.2 Hyperparameter Tuning

Tuning was conducted using nested stratified cross-validation (5 outer folds, 3 inner folds) and grid search.

- **Logistic Regression**: Tuned $C \in [0.01, 0.1, 1]$; best: $C = 0.1$ in 3/5 folds. *max_iter* was fixed at 1000 to ensure model convergence across all settings, and was not included in the grid search
- **Random Forest**: Tuned *n_estimators* ∈ [50, 100, 200], *max_depth* ∈ [2, 5, 10], *min_samples_leaf* ∈ [1, 2, 4]. Most frequently selected: *n_estimators* = 50 or 100, *max_depth* = 5, *min_samples_leaf* = 2.
- **KNN**: Tuned *n_neighbors* ∈ [1, 3, 5]; best: *n_neighbors* = 1 or 3. The distance metric was fixed as Euclidean distance, and weights were set to distance to stabilise the influence of varying *n_neighbors*.

Results

1. PCA and Oversampling / Undersampling

Although oversampling generally led to better performance across all models, the difference was most pronounced in KNN, where oversampling improved both accuracy and recall by around 8%. In contrast, logistic regression showed slightly worse performance (about 4%) on PCA-transformed data. Nevertheless, most metrics remained above 0.9 across all settings, and as shown in later sections, KNN performance improved significantly after hyperparameter tuning.

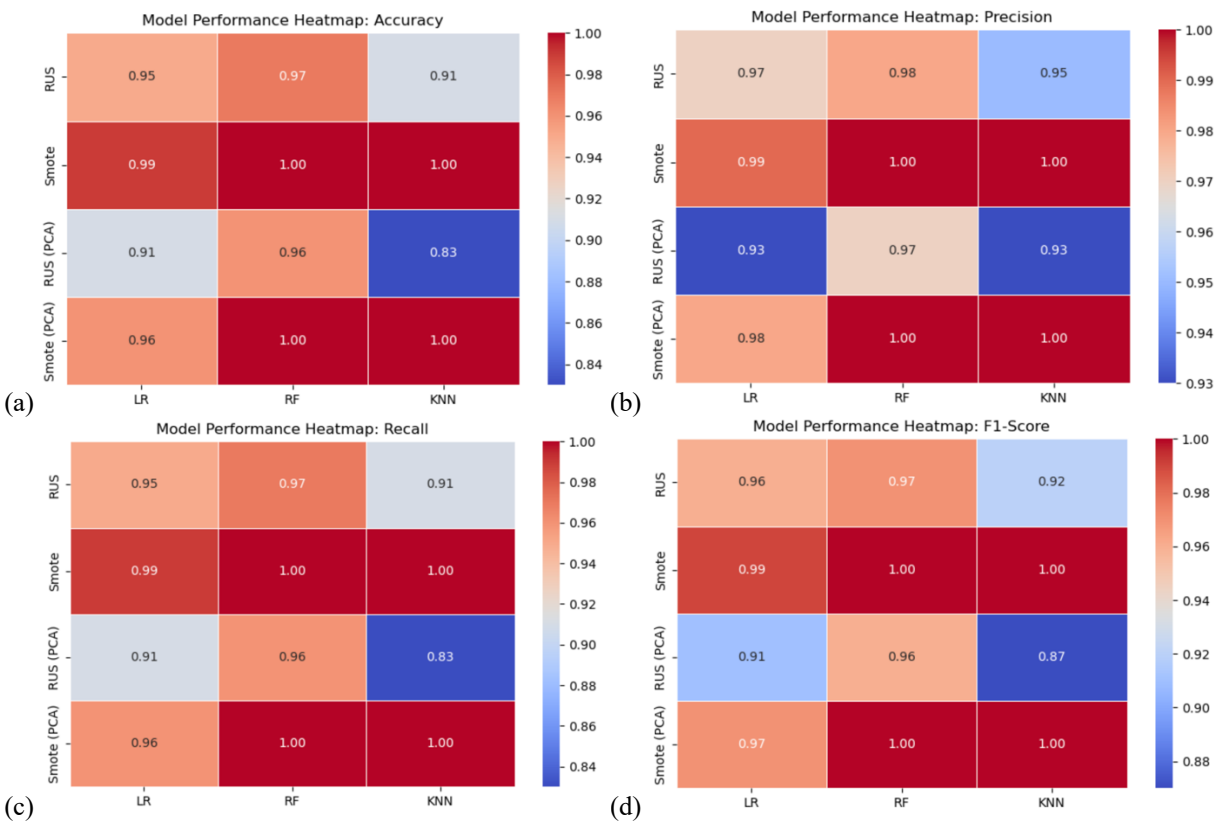

Fig2. Heatmap of Model Performance Across Four Training Datasets and Three Classifiers.
The four training datasets are: (1) oversampling (SMOTE) on processed features, (2) undersampling (random undersampling) on processed features, (3) oversampling on PCA-transformed features, and (4) undersampling on PCA-transformed features. The heatmap shows four subplots for evaluation metrics: (a) accuracy, (b) precision, (c) recall, and (d) F1 score.

2. Evaluation Metrics:

Following the imbalance handling results, tuning is performed on the undersampled PCA data. All models exhibited similar performance trends, with Logistic Regression and Random Forest achieving approximately 0.92 across all four metrics. However, KNN outperformed both LR and RF, reaching around 0.97 overall accuracy and a precision of 0.98.

LR and RF frequently misclassified Thing Speak and ARP poisoning instances, KNN greatly reduced these errors but showed a higher rate of misclassifying ARP poisoning as Metasploit Brute Force SSH.

- LR and RF misclassified ~750 instances (3%) of Thing Speak (normal pattern) as ARP poisoning. KNN reduced this to 29 (0.1%).
- LR and RF misclassified ~400 (1.6%) ARP poisoning as NMAP_UDP_SCAN. KNN made no such errors.
- KNN had more misclassifications of ARP poisoning as Metasploit Brute Force SSH (197 instances, 0.8%), higher than LR and RF.

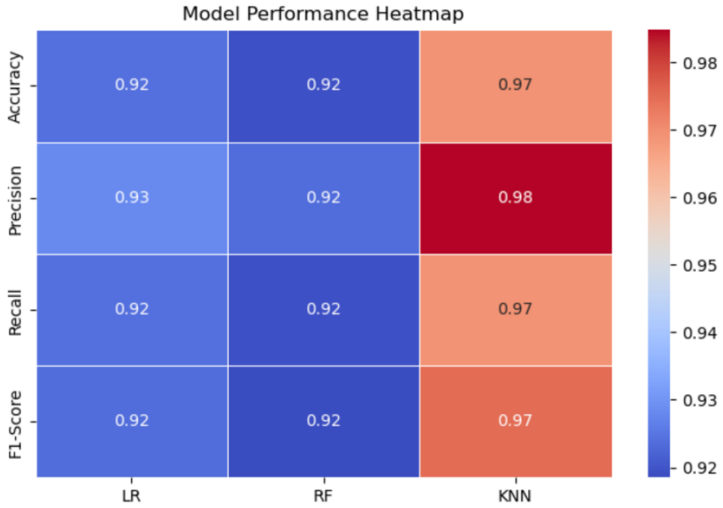

Fig3. Performance Heatmap of Tuned Classification Models.
KNN outperformed both LR and RF in all four metrics.

3. Learning Curves

LR and RF showed steadily improving validation accuracy with more training data, indicating further potential with larger datasets.

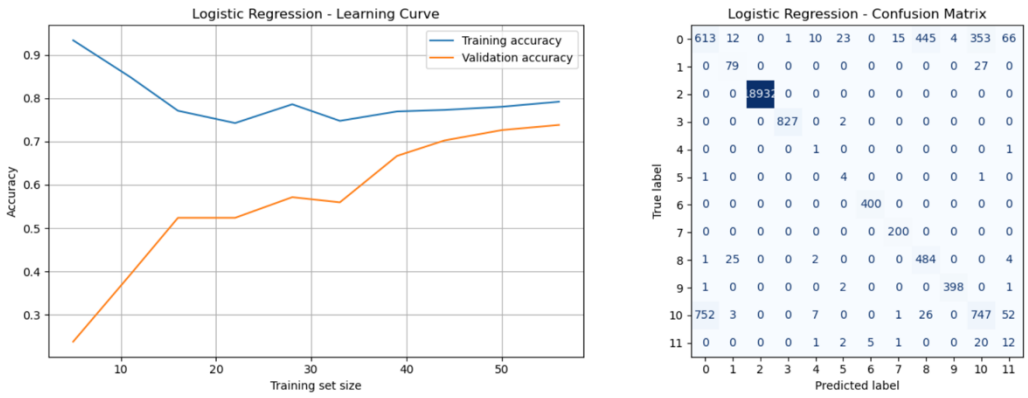

Fig4. Learning Curve (Left) and Confusion Matrix (Right) of the Tuned Logistic Regression Model.

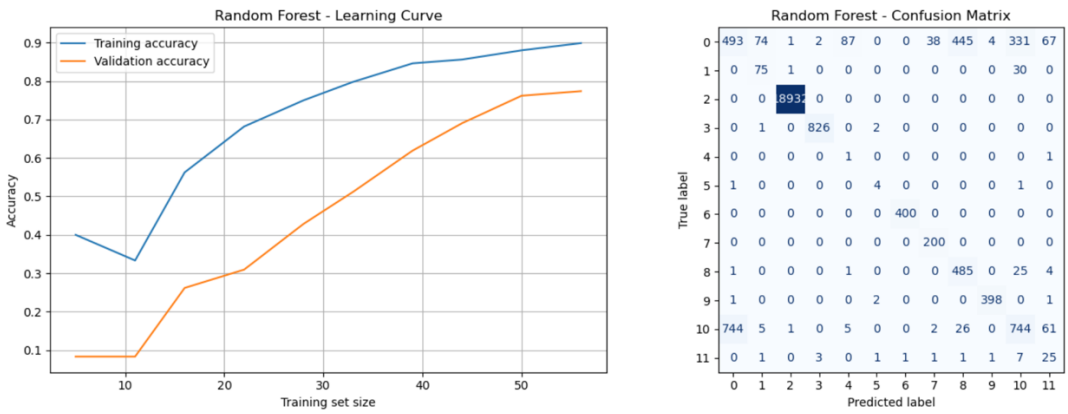

Fig5. Learning Curve (Left) and Confusion Matrix (Right) of the Tuned Random Forest Model.

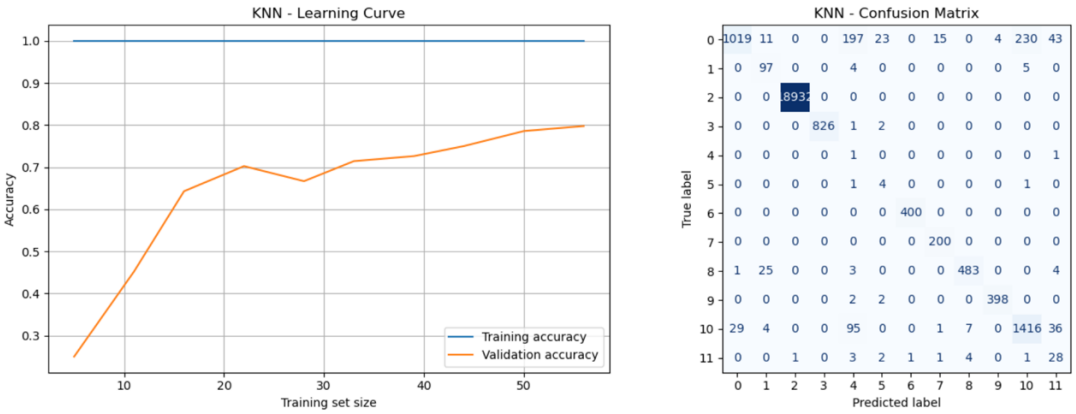

Fig6. Learning Curve (Left) and Confusion Matrix (Right) of the Tuned KNN Model.

Discussion

This project applied several machine learning models for multiclass classification of IoT network attacks, focusing on model performance, feature engineering, and handling data imbalance. **Undersampling** generally reduced performance, particularly in **KNN**, likely due to dimensionality reduction causing loss of important features and similar distributions among classes. However, **tuning hyperparameters**—especially using **smaller *n_neighbors***—improved KNN's results, highlighting the need for careful tuning on reduced datasets.

PCA significantly reduced training time and had minimal impact on Random Forest performance. Logistic Regression and KNN initially showed some performance decline, but KNN recovered notably after tuning, making PCA suitable for **real-time detection scenarios**.

Feature engineering was key: **TCP flag-based features** like SYN-ACK and RST ratios consistently showed high relevance. However, the dataset's **limited protocol-level details** hindered detection of certain attacks like ARP poisoning, which was often misclassified. **Class imbalance** further affected accuracy on rare attack types and may have led to underutilisation of directional features.

Overall, **KNN achieved the best performance** after tuning, consistent with literature. Logistic Regression and Random Forest showed promise for improvement with more balanced or larger datasets.

Conclusion

This study demonstrates that machine learning can be effectively applied to multiclass classification of IoT network attacks, especially when combined with appropriate feature engineering and data balancing techniques. While undersampling can harm model performance, proper tuning can mitigate this effect. PCA was shown to significantly enhance computational efficiency with minimal accuracy loss, making it suitable for real-time applications.

The project also highlights the critical role of TCP flag ratios in identifying IoT attack types and confirms that KNN, when tuned, outperforms other models in handling complex multiclass datasets, especially under imbalance conditions. However, the persistent misclassification of rare attack types, such as ARP poisoning, points to the need for richer protocol-level features and improved class representation in future datasets. Overall, the results suggest that, while existing models show promising performance, further improvements in dataset quality and balance are essential to support accurate and robust intrusion detection systems in IoT environments.

References

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CLARITY AND STRUCTURE (6): * The report follows an correct structure. * Citing at least one research paper - Yes COMPLETENESS AND CORRECTNESS (12): * Which feature construction and preprocessing (beyond simple methods) - Only simple methods used. * Three algorithms from high-bias/low-variance to low-bias/high-variance - LR, RF, and k-NN are considered! * Nested cross validation correctly implemented - 5-fold outside, and 3-fold inside are used for cross validation * Results presented (Loss with respect to number of iterations/epochs is not acceptable) - Several parameters are tuned for each model, and all selected values are not extreme ones. It reports Accuracy, Precision, Recall, and F1. CRITICAL ANALYSIS (7): * Proposed methods well motivated, advantages/disadvantes discussed - The reason behind the selection of three mentioned methods is missing. * Thorough and insightful analysis of why the proposed methods work/not work for the problem (beyond method A better than method B) - The report only connects model performance to feature extraction and overfitting control, lacking insightful analysis of the models' strengths and weaknesses. * Insightful discussion and analysis of alternative approaches and why they were not used (beyond method A better than method B) - No alternative methods is mentioned